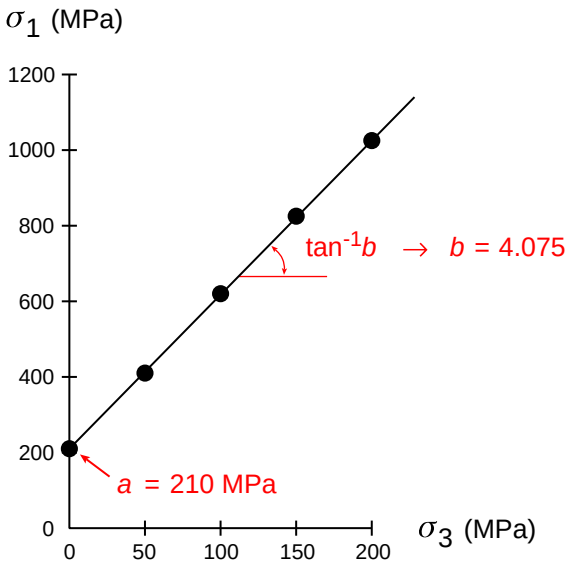
Brittle failure and the Mohr Circle (Solution)

The following data give the differential stresses (σ_1 - σ_3) supported at failure by a sandstone when loaded in axial compression at several different confining pressures (σ_3)

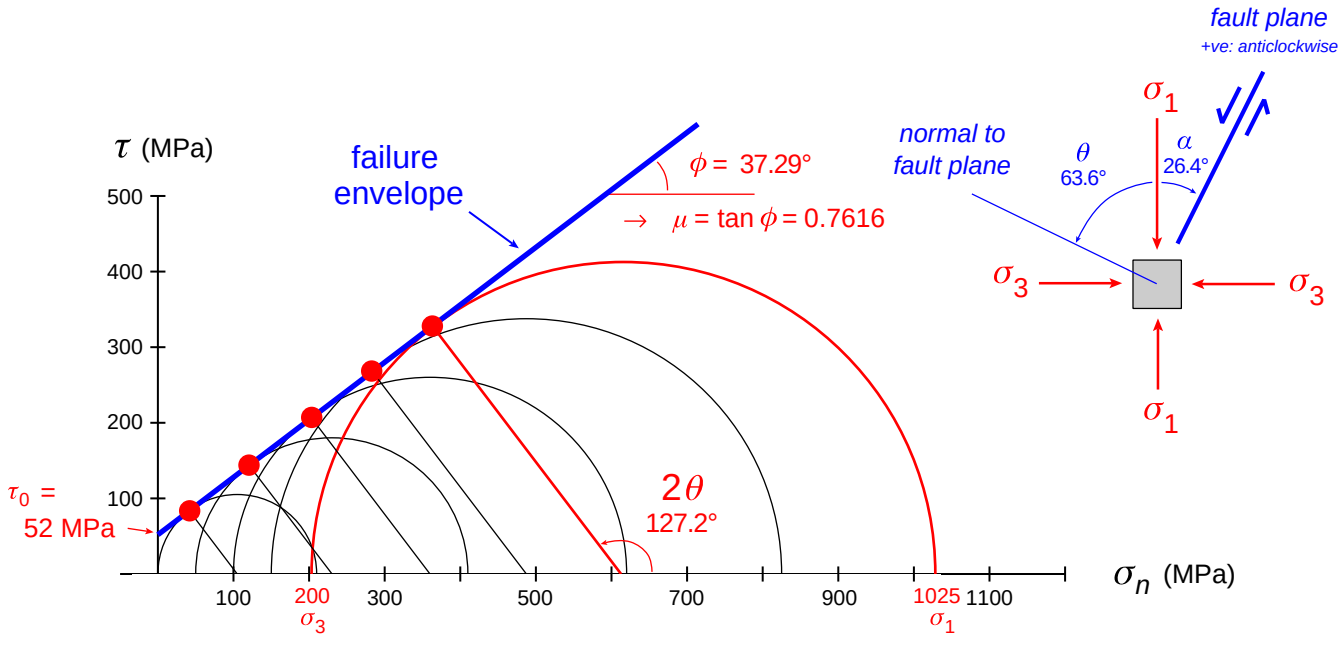
σ_3 (MPa)	σ_1 - σ_3 (MPa)
0	210
50	360
100	520
150	675
200	825

(a) Plot the data as σ_1 vs. σ_3 and determine the values of a and b in the failure criterion, $\sigma_1 = a + b\sigma_3$

σ_1 (MPa)	σ_3 (MPa)
210	0
410	50
620	100
825	150
1025	200



(b) Plot the data on a Mohr diagram and determine the values of cohesive strength (τ_0) and coefficient of internal friction (μ) in the failure criterion, $\tau = \tau_0 + \mu\sigma_n$. Determine the angle of the fault with respect to σ_1.



(c) OPTIONAL (at home): Starting with a diagram illustrating the geometry of the Mohr circle, show that

$$\sigma_1 = a + b\sigma_3$$

is equivalent to

$$\tau = \tau_0 + \sigma_n \tan\phi$$

where

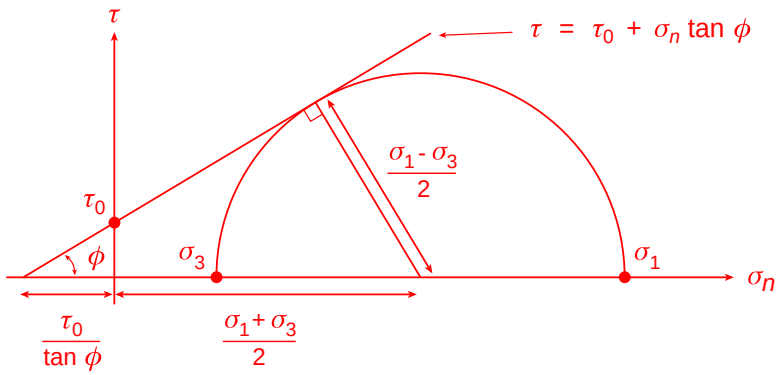
$$a = 2\tau_0 b^{½} \quad ; \quad b = (1 + \sin\phi) / (1 - \sin\phi)$$

You will find it helpful to make use of the trigonometrical identity $\cos^2\phi + \sin^2\phi = 1$ rewritten as

$$\cos^2\phi = 1 - \sin^2\phi = (1 + \sin\phi)(1 - \sin\phi)$$

Confirm these relationships using your answers in parts (a) and (b).

From the geometry of the Mohr circle



Hence from elementary trigonometry

$$\frac{\sigma_1 - \sigma_3}{2} = \left[\frac{\sigma_1 + \sigma_3}{2} + \frac{\tau_0}{\tan\phi}\right] \sin\phi$$

so that rearranging

$$\sigma_1 - \sigma_3 = \sigma_1 \sin\phi + \sigma_3 \sin\phi + 2\tau_0 \cos\phi$$

and so

$$\sigma_1 = \frac{2\tau_0 \cos\phi}{(1 - \sin\phi)} + \frac{(1 + \sin\phi)}{(1 - \sin\phi)} \sigma_3$$

Therefore

$$a = 2\tau_0 \cos\phi / (1 - \sin\phi) \quad ; \quad b = (1 + \sin\phi) / (1 - \sin\phi)$$

Now from the identity $\cos^2\phi + \sin^2\phi = 1$ then

$$\cos^2\phi = 1 - \sin^2\phi = (1 + \sin\phi)(1 - \sin\phi)$$

and so

$$\cos^2\phi / (1 - \sin\phi) = (1 + \sin\phi)$$

Dividing both sides by $(1 - \sin\phi)$

$$\cos^2\phi / (1 - \sin\phi)^2 = (1 + \sin\phi) / (1 - \sin\phi) = b$$

so that

$$a = 2\tau_0 b^{½} \quad ; \quad b = (1 + \sin\phi) / (1 - \sin\phi)$$

Check using answers in (a) and (b):

$a = 210 = 2 \times 52 \times 4.075^{½} = 2\tau_0 b^{½}$ Yes

$b = 4.075 = (1 + \sin 37.29) / (1 - \sin 37.29) = (1 + \sin\phi) / (1 - \sin\phi)$ Yes

(d) What would be the normal stress and resolved shear stress at failure on an optimally oriented fault formed in this sandstone at a depth of 3 km, if the fault was in an area of contractional tectonics? Assume that a principal stress is vertical. Note that the vertical stress, σ_V, is given by $\sigma_V = \rho_{rock} g z$, where ρ_{rock} = the average density of the overlying rock column = 2.3 gcm^{-3}, and g = the acceleration due to gravity = 10 ms^{-2}. When applying this formula for vertical stress make sure that you use consistent units. To do this note that the S.I. unit of stress is the Pascal:

$$1 \text{ Pa} = 1 \text{ Nm}^{-2} \quad \text{and that from } F = ma \quad 1 \text{ N} = 1 \text{ kgms}^{-2}$$

$$\sigma = \rho g z$$

$$\frac{[\text{N}]}{[\text{m}^2]} = \frac{[\text{kg}][\text{m}]}{[\text{s}^2]} \frac{1}{[\text{m}^2]} \quad \rightarrow \quad [\text{Pa}] = \frac{[\text{kg}]}{[\text{m}^3]} \frac{[\text{m}]}{[\text{s}^2]} [\text{m}]$$

Consistent units are: kg/m^3 for ρ, m/s^2 for g, m for z $\rightarrow$ N/m^2 = Pa

Conversion of g/cm^3 to kg/m^3

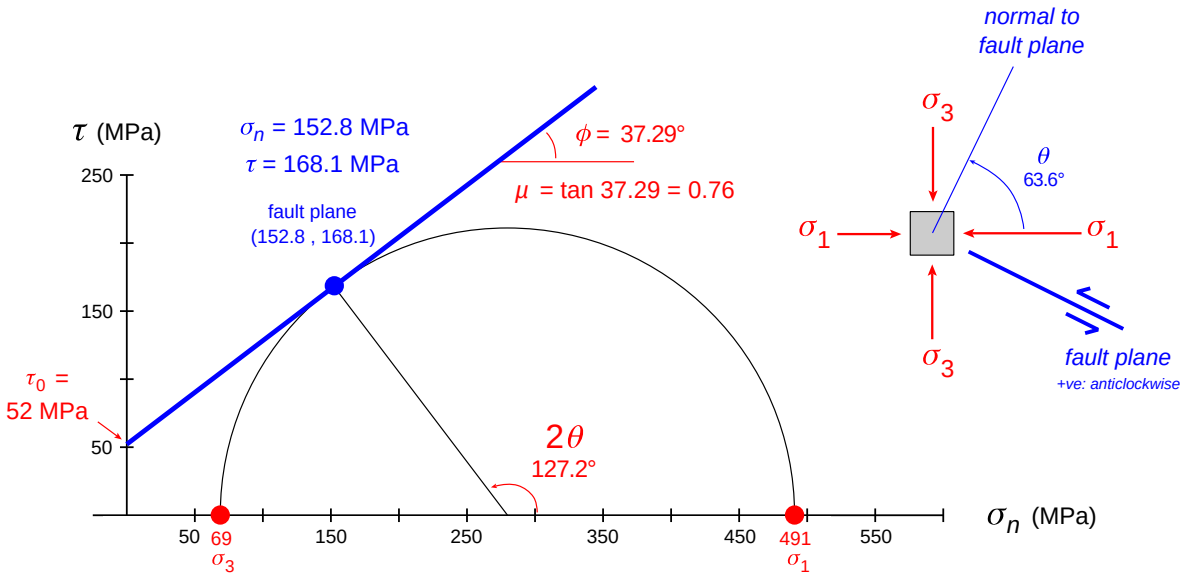
$$1 \frac{\text{g}}{\text{cm}^3} = \frac{0.001 \text{ kg}}{0.000001 \text{ m}^3} = 1000 \frac{\text{kg}}{\text{m}^3} \quad \rightarrow \quad 2.3 \text{ g/cm}^3 = 2300 \text{ kg/m}^3$$

The vertical stress (= σ_3 in an area of contractional tectonics) is given by

$$\sigma_3 = \sigma_V = \rho g z = 2300 \times 10 \times 3000 = 69 \text{ MPa}$$

Draw a circle with σ_3 = 69 MPa that just touches the failure envelope OR calculate first using

$$\sigma_1 = a + b\sigma_3 = 210 + (4.075 \times 69) = 491 \text{ MPa}$$



Or using formulae:

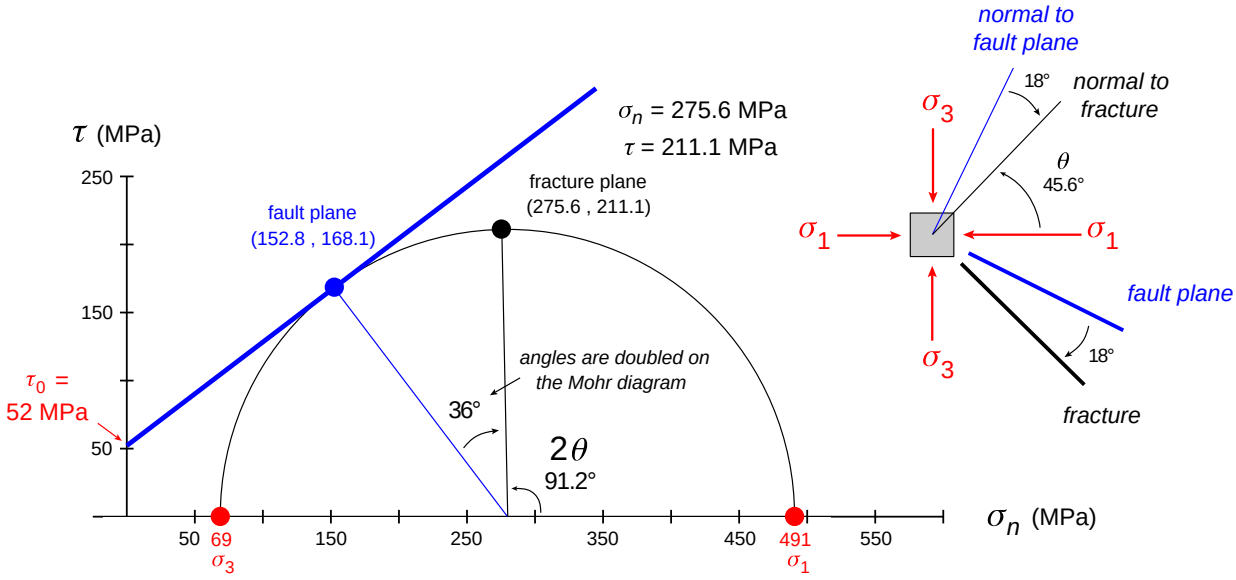
$$\tau = [(\sigma_1 - \sigma_3)/2] \sin 2\theta = [422/2] \sin 127.2 = 168.1 \text{ MPa}$$

$$\tau = \tau_0 + \mu\sigma_n = 52 + 0.76\sigma_n \quad \rightarrow \quad \sigma_n = (\tau - 52)/0.76 = 116.1/0.76 = 152.8 \text{ MPa}$$

Note that if you repeated this analysis for a region of extensional tectonics, then because now σ_1 is vertical, the rocks are very much weaker (a very much smaller diameter Mohr circle is needed if σ_1 = 69 MPa rather than σ_3).

(e) What would be the angle of dip of the fault in (d)? What would be the shear stress and normal stress acting on a fracture plane with the same dip direction as the fault but which dips 18° more steeply?

The angle between σ_1 (horizontal) and the normal to the fault is 63.6° and so the angle between σ_1 and the fault plane is 90° - 63.6° = 26.4°



Note that the shear stress on the fracture plane (211.1 MPa) is greater than on the fault plane but the fracture does not fail because the normal stress (275.6 MPa) is also greater.

(f) Repeat (d) but for the situation where there is a pore fluid (water) present that is connected up to the surface. Note that the normal stresses now need to be effective normal stresses.

The density of water is 1 gcm^{-3} and so the pore fluid pressure is given by

$$p_{fluid} = \rho_{water} g z = 1000 \times 10 \times 3000 = 30 \text{ MPa}$$

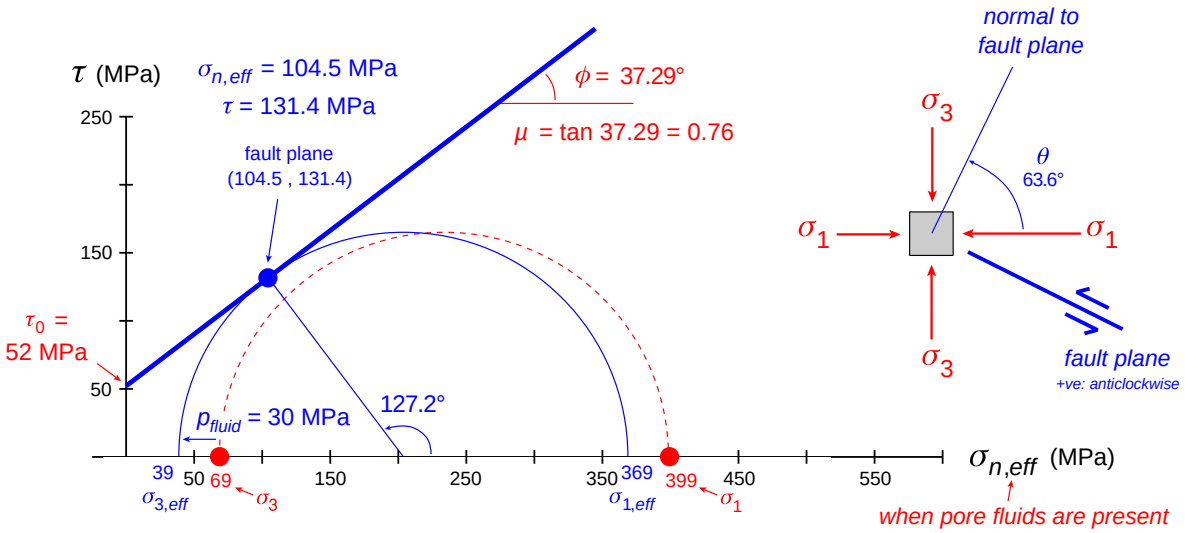
$\sigma_{3,eff} = \sigma_3 - p_{fluid} = 69 - 30 = 39$ MPa

$\sigma_{1,eff} = a + b\sigma_{3,eff} = 210 + (4.075 \times 39) = 369$ MPa

and so

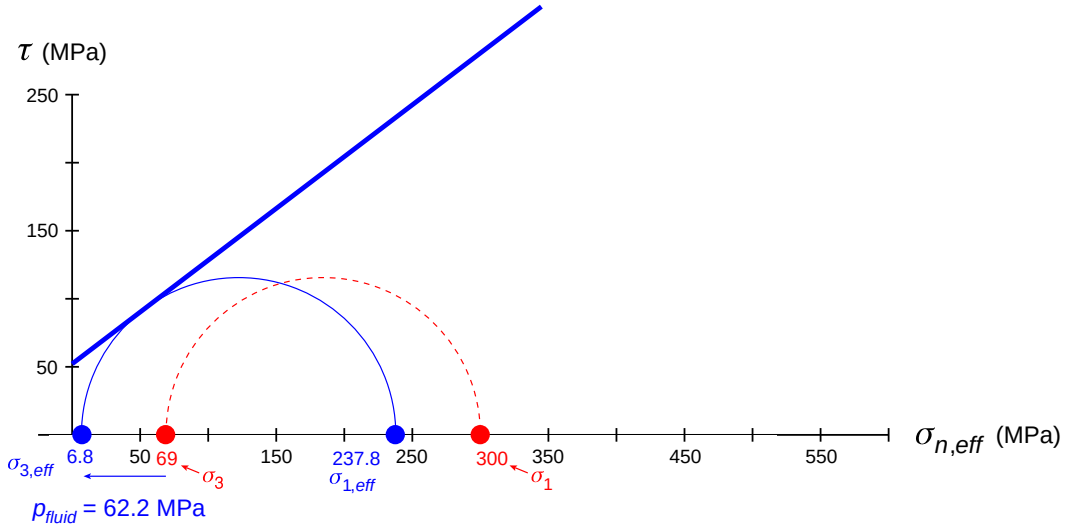
$\tau = [(\sigma_{1,eff} - \sigma_{3,eff}) / 2] \sin 2\theta = [330 / 2] \sin 127.2 = 131.4$ MPa

$\tau = \tau_0 + \mu\sigma_{n,eff} = 52 + 0.76\sigma_{n,eff} \rightarrow \sigma_{n,eff} = (\tau - 52) / 0.76 = 79.4 / 0.76 = 104.5$ MPa



Note that the rock is weaker (smaller $\sigma_1 - \sigma_3$) when there is a pore fluid present.

(g) If the value of the maximum principal stress at 3 km depth is 300 MPa what value of pore fluid pressure is required to cause failure?



The pore fluid pressure is given by the sideways shift of the Mohr circle that is required to get it to touch the failure envelope = 62.2 MPa.

Alternatively, this may be solved from the failure criterion. Working in effective stresses:

$\sigma_{1,\,eff} = a + b\sigma_{3,\,eff} \quad \rightarrow \quad \sigma_1 - p_{fluid} = a + b(\sigma_3 - p_{fluid})$

and so

$p_{fluid} = (a + b\sigma_3 - \sigma_1) / (b - 1)$

From (a) we have: a = 210 MPa ; b = 4.075
From (d) at 3 km depth we have σ_3 = 69 MPa, and so with σ_1 = 300 MPa, then

$p_{fluid} = 62.2$ MPa

The petroleum industry commonly uses the pore pressure parameter, λ, to assess pore pressures. λ is defined,

$\lambda \equiv p_{fluid} / \sigma_V$

For normally pressured pore fluids (fluids connected all the way up to the surface) we see from (f) that

$\lambda \equiv p_{fluid} / \sigma_V = 30 / 69 = 0.43$

A value of λ greater than this means that the fluids are over-pressured (liable to "erupt" if breached by a borehole). In scenario (g) the fluids are highly overpressured:

$\lambda \equiv p_{fluid} / \sigma_V = 62.2 / 69 = 0.90$

The maximum value of $\lambda \approx 1$ because is $p_{fluid} > \sigma_V$ then the rocks are in tension and rocks are very weak in tension.